Learning as Inference

DD2421

André Pereira
(slides heavily adapted from Bob Sturm)

Outline

1. Introduction
   - Probabilistic Classification and Regression
   - Discriminative vs Generative Models
   - Parametric vs Non-parametric Inference
2. Maximum Likelihood (ML) Estimation
   - Regression
   - Classification
3. Special Cases
   - Naïve Bayes Classifier
   - Logistic Regression

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Probabilistic Classification and Regression

- In both cases we compute the posterior

$$\mathrm{p}(y \mid \mathbf{X} = \mathbf{x}) = \frac{\mathrm{p}(\mathbf{x} \mid \mathrm{Y} = y)\,\mathrm{p}(\mathrm{Y} = y)}{\mathrm{p}(\mathbf{X} = \mathbf{x})}$$

- Classification: y is discrete
- Regression: y is continuous

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Until now we assumed we knew:

- $\mathrm{p}(Y = y) \equiv \mathrm{p}(y) \leftarrow$ *Prior*
- $\mathrm{p}(x|Y = y) \equiv \mathrm{p}(x|y) \leftarrow$ *Likelihood*
- $\mathrm{p}(X = x) \equiv \mathrm{p}(x) \leftarrow$ *Evidence*

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How can we obtain these distributions from data?

Learning as Inference

Given:

- the training data $\mathcal{D} = \{(\mathrm{x}, y)_1, (\mathrm{x}, y)_2, \ldots, (\mathrm{x}, y)_N\}$
- a new observation x

Estimate the posterior probability of y:

$$\mathrm{p}(y|\mathrm{x}, \mathcal{D})$$

Discriminative vs Generative Models

Discriminative modeling:

- This models $\mathrm{p}(y|\mathrm{x}, \mathcal{D})$ directly
- example: Logistic regression

Generative modeling:

- This models $\mathrm{p}(\mathrm{x}|\mathrm{y}, \mathcal{D})$
- example: Naive Bayes

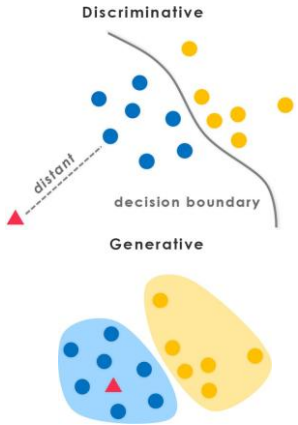


Figure from Nguyen *et al.* 2015.

Parametric vs Non-parametric Inference

$$\mathrm{p}(y|\mathrm{x}) = \mathrm{p}\,(y|\mathrm{x}, \theta)$$

The distribution is characterized by parameters θ.

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Parametric Inference:

- Estimate θ using $\mathcal{D}$
- Compute $\mathrm{p}(y|\mathbf{x}, \hat{\theta})$ to make inference.

Learning corresponds to estimating θ

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Non-Parametric Inference:

- Estimate $\mathrm{p}(\theta|\mathcal{D})$
- Compute $\mathrm{p}(y|\mathbf{x}, \mathcal{D})$ from $\mathrm{p}(y|\mathbf{x}, \theta, \mathcal{D})\ \mathrm{p}(\theta|\mathcal{D})$ by marginalizing out θ

The number of parameters can grow with the data!

Three Approaches

Parametric inference:

- Maximum Likelihood (ML) Estimation (today)
- Maximum A Posteriori (MAP) Estimation (next time)

Non-parametric inference:

- Bayesian methods (a little today and the rest next time)

Fundamental Assumption: i.i.d.

Observations are independent and identically distributed (i.i.d.):

$$\mathcal{D} = \{\boldsymbol{o}_1, \ldots, \boldsymbol{o}_N\}, \boldsymbol{o}_i = (\mathbf{x}, y)_i$$

The likelihood of the whole data set can be factorized:

$$\mathrm{p}(\mathcal{D}) = \mathrm{p}(\boldsymbol{o}_1, \ldots, \boldsymbol{o}_N) = \prod_{i=1}^{N} \mathrm{p}(\boldsymbol{o}_i)$$

Taking the log creates the log-likelihood

$$\log \mathrm{p}(\mathcal{D}) = \sum_{i=1}^{N} \log \mathrm{p}(\boldsymbol{o}_i)$$

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Maximum Likelihood (ML) Estimate

$$\mathrm{p}(\boldsymbol{x}|y) \equiv \mathrm{p}(\boldsymbol{x}|y,\theta) \quad \text{or} \quad \mathrm{p}(y|\boldsymbol{x}) \equiv \mathrm{p}(y|\boldsymbol{x},\theta)$$

Find the parameter values that make the data most likely.

- *ML optimality* is defined as maximizing the likelihood of D:

$$\theta_{\mathrm{ML}} = \arg\max_{\theta} \mathrm{p}(\mathcal{D}|\theta) = \arg\max_{\theta} \log \mathrm{p}(\mathcal{D}|\theta)$$

- We can then approximate distributions given the data:

$$\mathrm{p}(\mathbf{x}|y,\mathcal{D}) \approx \mathrm{p}(\mathbf{x}|y,\theta_{ML}) \quad \text{or} \quad \mathrm{p}(y|\mathbf{x},\mathcal{D}) \approx \mathrm{p}(y|\mathbf{x},\theta_{ML})$$

Probabilistic Linear Regression

Model (deterministic):

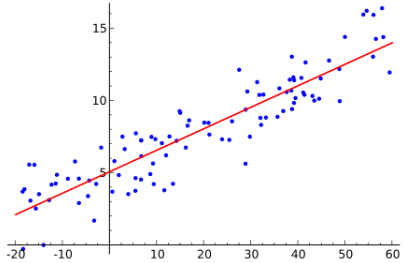
$$y = \boldsymbol{w}^T\mathbf{x} + \boldsymbol{\epsilon}$$

But now:

$$\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2)$$

Therefore:

$$Y \sim \mathcal{N}\left(\mu_Y(\mathbf{x}), \sigma_Y^2(\mathbf{x})\right) = \mathcal{N}(\boldsymbol{w}^T\mathbf{x}, \sigma^2)$$



Learning: find **w** that maximizes $\mathrm{p}(y|\mathbf{x}, \boldsymbol{w}, \sigma^2)$

Maximize the posterior directly $\Rightarrow$ discriminative method

MLE for Probabilistic Linear Regression

$$
\begin{aligned}
\log \mathrm{p}(y|\mathbf{x}, \boldsymbol{w}, \sigma^2) &= \log \prod_i \mathrm{p}(\, y_i \mid \mathbf{x_i}, \mathbf{w}, \sigma^2\,) \\
&= \sum_i \log \mathrm{p}(\, y_i \mid \mathbf{x_i}, \mathbf{w}, \sigma^2\,) \\
&= \sum_i \log \left[\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_i - \mathbf{w}^\top \mathbf{x_i})^2}{2\sigma^2} \right) \right] \\
&= \sum_i \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{(y_i - \mathbf{w}^\top \mathbf{x_i})^2}{2\sigma^2} \right]
\end{aligned}
$$

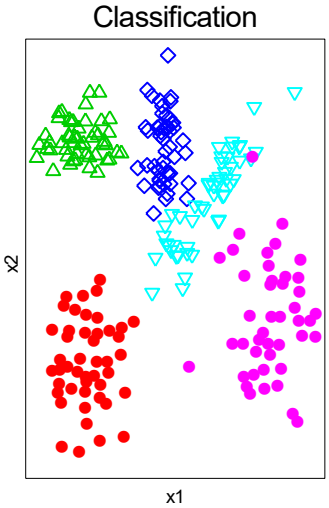
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\end{aligned}
$$

$$
\arg\max_{\boldsymbol{w}} \mathrm{p}(y|\mathbf{x}, \boldsymbol{w}, \sigma^2) = \arg\min_{\boldsymbol{w}} \sum_i (y_i - \mathbf{w}^\top \mathbf{x_i})^2
$$

NEAT-O! Choosing parameters that maximize $\mathrm{p}(y|\mathbf{x}, \boldsymbol{w}, \sigma^2)$ ≡ minimizing mean square error! (in this case)

MLE for Classification

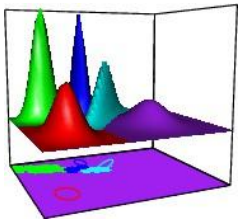


features: $\mathbf{x} \in \mathbb{R}^d$

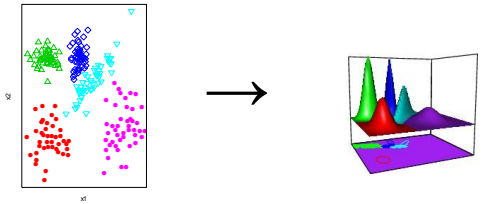
class: $y \in \{y_1, \ldots, y_K\}$

$$k_{MAP} = \arg\max_{k} \mathrm{p}(y_k|\mathbf{x})$$

$$= \arg\max_{k} \mathrm{p}(\mathrm{x}|y_k)\,\mathrm{p}(y_k)$$

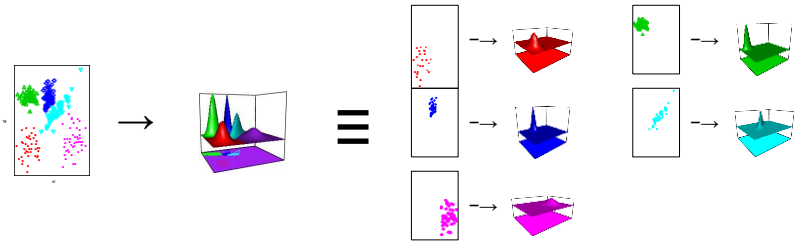


Assumption: Class Independence



samples from class i do not influence estimate for class j, $i \neq j$

Assumption: Class Independence



- distribution of $\mathbf{x}$ for class y_k is the likelihood $\Pr(\boldsymbol{x}|\theta_k)$
- in the following, we drop the class index k and write $\Pr(\boldsymbol{x}|\theta)$
- also we call $\mathcal{D} = \{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N\}$ the set of data point belonging to a single class y_k

ML estimation of Gaussian mean

$$X \sim \mathcal{N}(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right], \text{ with } \theta = \{\mu, \sigma^2\}$$

Log-likelihood of data (**i**ndependent **i**dentical **d**istribution samples):

$$\log \mathrm{p}(\mathcal{D} \mid \theta) = \sum_{n=1}^{N} \log \mathcal{N}(x_n \mid \mu, \sigma^2)$$

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$$0 = \frac{d}{d\mu} \log \mathrm{p}(\mathcal{D} \mid \theta)$$

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$$\mu_{ML} = \frac{1}{N} \sum_{n=1}^{N} x_n$$

ML estimation of Gaussian parameters

$$\mu_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} x_n$$

$$\sigma^2_{\mathrm{ML}} = \frac{1}{N}\sum_{n=1}^{N} (x_n - \mu_{\mathrm{ML}})^2$$

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- This is the same result as minimizing the sum of square errors!
- but now out assumptions are explicit (i.e., how the data is distributed)
- This estimate of the variance is *biased*, i.e., $\mathbb{E}[\sigma^2_{ML}] - \sigma^2 \neq 0$. The unbiased ML estimate is

$$\sigma'^2_{ML} = \frac{1}{N-1}\sum_{n=1}^{N} (x_n - \mu_{ML})^2$$

MLE with Discrete Variables

Will I go and play orienteering given the forecast?

$$x \in \{\text{sunny}, \text{overcast}, \text{rainy}\}$$
$$y \in \{\text{yes}, \text{no}\}$$

$$X \sim ?$$
$$Y \sim ?$$
$$X|Y \sim ?$$
$$Y|X \sim ?$$

Training data

n example	x_n outlook	y_n play	n example	x_n outlook	y_n play
1	sunny	no	8	sunny	no
2	sunny	no	9	sunny	yes
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$$Y \sim \text{Bernoulli}(\lambda)$$
$$X|Y \sim \text{Cat}(\lambda'_1, \lambda'_2, \lambda'_3)$$
$$Y|X \sim \text{Bernoulli}(\lambda')$$

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MLE: Bernoulli

$$p(y) = \begin{cases} \lambda, & \text{if } x = 1 \\ 1 - \lambda, & \text{if } x = 0 \end{cases}$$

1. compute (log) likelihood of the data $\mathrm{p}(\mathcal{D}|\lambda)$
2. find λ_{ML} that optimizes $\mathrm{p}(\mathcal{D}|\lambda)$

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Likelihood of the data
(n=number of yes in D, N=number of examples):

$$\mathrm{p}(\mathcal{D} \mid \lambda) = \prod_n \mathrm{p}(y_n \mid \lambda) = \prod_{n \text{ s.t. } y=\text{yes}} \lambda \prod_{n \text{ s.t. } y=\text{no}} (1-\lambda)$$

$$= \lambda^n (1-\lambda)^{N-n}$$

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$$\log \mathrm{p}(\mathcal{D} \mid \lambda) = n \log \lambda + (N-n) \log(1-\lambda)$$

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$$\log \mathrm{p}(\mathcal{D} \mid \lambda) = n \log \lambda + (N-n) \log(1-\lambda)$$

$$\frac{d}{d\lambda} \log \mathrm{p}(\mathcal{D} \mid \lambda) = \frac{n - N\lambda}{\lambda(1-\lambda)} = 0$$

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$$= \lambda^n (1-\lambda)^{N-n}$$

$$\log \mathrm{p}(\mathcal{D} \mid \lambda) = n \log \lambda + (N-n) \log(1-\lambda)$$

$$\frac{d}{d\lambda} \log \mathrm{p}(\mathcal{D} \mid \lambda) = \frac{n - N\lambda}{\lambda(1-\lambda)} = 0 \Leftrightarrow \lambda_{\mathrm{ML}} = \frac{n}{N}$$

MLE with Discrete Variables

Will I go and play orienteering given the forecast?

$$x \in \{\text{sunny}, \text{overcast}, \text{rainy}\}$$
$$y \in \{\text{yes}, \text{no}\}$$

$$Y \sim \text{Bernoulli}(\lambda)$$
$$\lambda_{\text{ML}} = \frac{9}{14}$$

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MLE: Categorical

Similar derivation:

$$\lambda_{k,\mathrm{ML}} = \frac{n_k}{N}$$

where n_k is the number of examples of the kth category

$$X \sim \mathrm{Cat}\left(\lambda_{sunny}, \lambda_{\mathrm{overcast}}, \lambda_{\mathrm{rainy}}\right)$$

$$\lambda_{k,\mathrm{ML}} = \left\{\frac{5}{14}, \frac{4}{14}, \frac{5}{14}\right\}$$

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$$X \sim \mathrm{Cat}\left(\lambda_{sunny}, \lambda_{\mathrm{overcast}}, \lambda_{\mathrm{rainy}}\right)$$

$$\lambda_{k,\mathrm{ML}} = \left\{\frac{5}{14}, \frac{4}{14}, \frac{5}{14}\right\}$$

$$X \mid Y \sim \mathrm{Cat}(\lambda'_1, \ldots, \lambda'_k)$$

Training data

n example	x_n outlook	y_n play	n example	x_n outlook	y_n play
1	sunny	no	8	sunny	no
2	sunny	no	9	sunny	yes
3	overcast	yes	10	rainy	yes
4	rainy	yes	11	sunny	yes
5	rainy	yes	12	overcast	yes
6	rainy	no	13	overcast	yes
7	overcast	yes	14	rainy	no

MLE: Categorical

Similar derivation:

$$\lambda_{k,\mathrm{ML}} = \frac{n_k}{N}$$

where n_k is the number of examples of the kth category

$$X \sim \mathrm{Cat}\left(\lambda_{sunny}, \lambda_{\mathrm{overcast}}, \lambda_{\mathrm{rainy}}\right)$$

$$\lambda_{k,\mathrm{ML}} = \left\{\frac{5}{14}, \frac{4}{14}, \frac{5}{14}\right\}$$

$$X \mid Y \sim \mathrm{Cat}(\lambda'_1, \ldots, \lambda'_k)$$

$$\lambda'_{\mathrm{ML}}(\mathrm{yes}) = \left\{\frac{2}{9}, \frac{4}{9}, \frac{3}{9}\right\}$$

$$\lambda'_{\mathrm{ML}}(\mathrm{no}) = \left\{\frac{3}{5}, 0, \frac{2}{5}\right\}$$

Training data

n example	x_n outlook	y_n play	n example	x_n outlook	y_n play
1	sunny	no	8	sunny	no
2	sunny	no	9	sunny	yes
3	overcast	yes	10	rainy	yes
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But . . . will I play orienteering given a rainy outlook?

$$\mathrm{p}(y = \text{yes} \mid \text{outlook} = \text{rainy}) = \frac{\mathrm{p}(\text{outlook} = \text{rainy} \mid y = \text{yes})\,\mathrm{p}(y = \text{yes})}{\mathrm{p}(\text{outlook} = \text{rainy})}$$

$$= \frac{\frac{3}{9}\frac{9}{14}}{\frac{5}{14}} = \frac{3}{5}$$

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$$\mathrm{p}(y = \text{no} \mid \text{outlook} = \text{rainy}) = \frac{\mathrm{p}(\text{outlook} = \text{rainy} \mid y = \text{no})\,\mathrm{p}(y = \text{no})}{\mathrm{p}(\text{outlook} = \text{rainy})}$$

$$= \frac{\frac{2}{5}\frac{5}{14}}{\frac{5}{14}} = \frac{2}{5}$$

Then

$$y_{\text{MAP}} = \arg\max_{y} \Pr(y \mid \text{outlook} = \text{rainy}) = \text{yes} \quad (3/5 > 2/5)$$

But . . . will I play orienteering given a rainy outlook?

$$\mathrm{p}(\, y = \text{yes} \mid \text{outlook} = \text{rainy}\,) = \frac{\mathrm{p}(\,\text{outlook} = \text{rainy} \mid y = \text{yes}\,)\,\mathrm{p}(y = \text{yes})}{\mathrm{p}(\text{outlook} = \text{rainy})}$$

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$$\mathrm{p}(\, y = \text{no} \mid \text{outlook} = \text{rainy}\,) = \frac{\mathrm{p}(\,\text{outlook} = \text{rainy} \mid y = \text{no}\,)\,\mathrm{p}(y = \text{no})}{\mathrm{p}(\text{outlook} = \text{rainy})}$$

$$= \frac{\frac{2}{5}\frac{5}{14}}{\frac{5}{14}} = \frac{2}{5}$$

Then

$$y_{\text{MAP}} = \arg\max_y \Pr(\, y \mid \text{outlook} = \text{rainy}\,) = \text{yes} \quad (3/5 > 2/5)$$

$$y_{\text{ML}} = \arg\max_y \Pr(\,\text{outlook} = \text{rainy} \mid y\,) = \text{no} \quad (2/5 > 3/9)$$

Source of confusion

Maximum a Posteriori (MAP) and Maximum Likelihood (ML) classification are *different*:

$$y_{\text{MAP}} = \arg\max_y P\left(y \mid \mathbf{x}, \theta_{\text{ML}}\right)$$

$$y_{\text{ML}} = \arg\max_y P\left(\mathbf{x} \mid y, \theta_{\text{ML}}\right)$$

even with parameters θ estimated with the ML *optimality criterion*:

$$\theta_{\text{ML}} = \arg\max_\theta P\left(\mathcal{D} \mid y, \theta\right) = \arg\max_\theta \prod_n P(\mathbf{x_n} \mid y_n, \theta)$$

NB: ML *parameter* estimation is not ML regression/classification.

Outline

1. Introduction
   - Probabilistic Classification and Regression
   - Discriminative vs Generative Models
   - Parametric vs Non-parametric Inference
2. Maximum Likelihood (ML) Estimation
   - Regression
   - Classification
3. Special Cases
   - Naïve Bayes Classifier
   - Logistic Regression

Problem: Curse of Dimensionality

n	$\mathbf{x}_n$				y_n
example	outlook	temperature	humidity	windy	play
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
3	overcast	hot	high	false	yes
4	rainy	mild	high	false	yes
5	rainy	cool	normal	false	yes
6	rainy	cool	normal	true	no
7	overcast	cool	normal	true	yes
8	sunny	mild	high	false	no
9	sunny	cool	normal	false	yes
10	rainy	mild	normal	false	yes
11	sunny	mild	normal	true	yes
12	overcast	mild	high	true	yes
13	overcast	hot	normal	false	yes
14	rainy	mild	high	true	no

difficult to model $p(outlook, temperature, humidity, windy|play)$

Problem: Curse of Dimensionality

- Volume of feature space exponential in number of features.
- More features $\Rightarrow$ potential for better description of the objects but …
- … $\Rightarrow$ need more and more data to model $p(x, y)$ well

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Approximation: **Naïve Bayes classifier**

- All features (dimensions) regarded as conditionally independent.
- Instead of modelling one *D*-dimensional distribution: $p(outlook, temperature, humidity, windy|play)$ model *D* one-dimensional distributions: $p(outlook|play)$, $p(temperature|play)$, $p(humidity|play)$, $p(windy|play)$

Naïve Bayes Classifier

- **x** is a vector $(x_1, \ldots, x_D)$ of attribute or feature values.
- Let $Y = \{1, 2, \ldots, K\}$ be the set of possible classes.
- MAP classification is

$$y_{MAP} = \arg\max_{y \in \mathcal{Y}} \Pr(y \mid x_1, \ldots, x_D) = \arg\max_{y \in \mathcal{Y}} \frac{\mathrm{p}(x_1, \ldots, x_D \mid y)\ \mathrm{p}(y)}{\mathrm{p}(x_1, \ldots, x_D)}$$

$$= \arg\max_{y \in \mathcal{Y}} \mathrm{p}(x_1, \ldots, x_D \mid y)\ \mathrm{p}(y)$$

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$$= \arg\max_{y \in \mathcal{Y}} \mathrm{p}(x_1, \ldots, x_D \mid y)\ \mathrm{p}(y)$$

- Naïve Bayes assumption:

$$\mathrm{p}(x_1, \ldots, x_D \mid y) = \prod_{d=1}^{D} \mathrm{p}(x_d \mid y)$$

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- Naïve Bayes assumption:

$$\mathrm{p}(x_1, \ldots, x_D \mid y) = \prod_{d=1}^{D} \mathrm{p}(x_d \mid y)$$

- MAP classification with Naïve Bayes:

$$y_{MAP} = \arg\max_{y \in \mathcal{Y}} \mathrm{p}(y) \prod_{d=1}^{D} \mathrm{p}(x_d \mid y)$$

Naïve Bayes Classifier

$$y_{MAP} = \arg\max_{y \in \mathcal{Y}} \mathrm{p}(y) \prod_{d=1}^{D} \mathrm{p}(x_d \mid y)$$

Naïve Bayes is one of the most common learning methods.
When to use:

- Moderate or large training set available.
- Feature dimensions are conditionally independent given class (or at least reasonably independent, still works with a little dependence).

Successful applications:

- Medical diagnoses (symptoms independent)
- Classification of text documents (words independent)

Example: Play Orienteering?

Question: Will I go and play orienteering given the forecast?

My measurements:

- **outlook** $\in \{sunny, overcast, rainy\}$,
- **temperature** $\in \{hot, mild, cool\}$,
- **humidity** $\in \{high, normal\}$,
- **windy** $\in \{false, true\}$.

Possible decisions: $y \in \{yes, no\}$

Example: Play Orienteering?

What I did in the past:

n	$\mathbf{x}_n$				y_n
example	outlook	temperature	humidity	windy	play
1	sunny	hot	high	false	no
2	sunny	hot	high	true	no
3	overcast	hot	high	false	yes
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Example: Play Orienteering?

Counts of when I played orienteering (did not play)

Outlook			**Temperature**			**Humidity**		**Windy**	
sunny	overcast	rain	hot	mild	cool	high	normal	false	true
2 (3)	4 (0)	3 (2)	2 (2)	4 (2)	3 (1)	3 (4)	6 (1)	6 (2)	3 (3)

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Prior of whether I played orienteering or not

Counts:

Play	
yes	no
9	5

Prior Probabilities:

Play	
yes	no
$\frac{9}{14}$	$\frac{5}{14}$

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Counts of when I played orienteering (did not play)

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Counts:

Play	
yes	no
9	5

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yes	no
$\frac{9}{14}$	$\frac{5}{14}$

Likelihood of attribute when orienteering played $Pr(x_i \mid \text{y=yes})(Pr(x_i \mid \text{y=no}))$

Outlook			Temperature			Humidity		Windy	
sunny	overcast	rain	hot	mild	cool	high	normal	false	true
$\frac{2}{9}$ $\left(\frac{3}{5}\right)$	$\frac{4}{9}$ $\left(\frac{0}{5}\right)$	$\frac{3}{9}$ $\left(\frac{2}{5}\right)$	$\frac{2}{9}$ $\left(\frac{2}{5}\right)$	$\frac{4}{9}$ $\left(\frac{2}{5}\right)$	$\frac{3}{9}$ $\left(\frac{1}{5}\right)$	$\frac{3}{9}$ $\left(\frac{4}{5}\right)$	$\frac{6}{9}$ $\left(\frac{1}{5}\right)$	$\frac{6}{9}$ $\left(\frac{2}{5}\right)$	$\frac{3}{9}$ $\left(\frac{3}{5}\right)$

Example: Play Orienteering?

Inference: Use the learnt model to classify a new instance.

New instance:

$$\mathbf{x} = (sunny, cool, high, true)$$

Apply Naïve Bayes Classifier:

$$y_{MAP} = \arg \max_{y \in \{\text{yes,no}\}} \mathrm{p}(y) \prod_{\mathrm{i}=1}^{4} \mathrm{p}(x_{\mathrm{i}} \mid y)$$

$$p(\text{yes})\, p(\text{sunny} \mid \text{yes})\, p(\text{cool} \mid \text{yes})\, p(\text{high} \mid \text{yes})\, p(\text{true} \mid \text{yes}) = \frac{9}{14} \times \frac{2}{9} \times \frac{3}{9} \times \frac{3}{9} \times \frac{3}{9} = .005$$

$$p(\text{no})\, p(\text{sunny} \mid \text{no})\, p(\text{cool} \mid \text{no})\, p(\text{high} \mid \text{no})\, p(\text{true} \mid \text{no}) = \frac{5}{14} \times \frac{3}{5} \times \frac{1}{5} \times \frac{4}{5} \times \frac{3}{5} = .021$$

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$\Rightarrow$ y_{MAP} = no

Naïve Bayes: Independence Violation

- Conditional independence assumption:

$$\mathrm{p}(x_1, \ldots, x_D \mid y) = \prod_{d=1}^{D} \mathrm{p}(x_d \mid y)$$

  often violated – but it works surprisingly well anyway!

- Since dependencies ignored, naïve Bayes posteriors often unrealistically close to 0 or 1.

  Different attributes say the same thing to a higher degree than we expect as they are correlated in reality.

Naïve Bayes: Estimating Probabilities

- **Problem:** What if none of the training instances with target value y have attribute x_i? Then

$$p(x_i \mid y) = 0 \quad \Longrightarrow \quad p(y) \prod_{i=1}^{D} p(x_i \mid y) = 0$$

Naïve Bayes: Estimating Probabilities

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$$p(x_i \mid y) = 0 \quad \Longrightarrow \quad p(y) \prod_{i=1}^{D} p(x_i \mid y) = 0$$

- **Simple solution:** add pseudocounts to all counts so that no count is zero

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- **Simple solution:** add pseudocounts to all counts so that no count is zero
- This is a form of regularization or smoothing

Logistic Regression

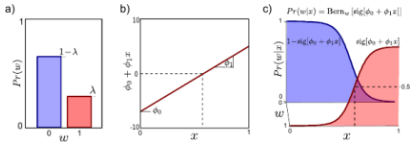


Figure from Prince (Ch. 9)

- Binary classification problem: $y \in \{0, 1\}$ treated as a regression problem: $\mathbf{x} \rightarrow \lambda$ $(Bernoulli\ param.)$

$$Y \mid \mathbf{X} \sim \text{Bernoulli}\big(\lambda(\mathbf{x})\big)$$

$$\Pr(\, y \mid \mathbf{x}\,) = \lambda(\mathbf{x})^{y}\big(1 - \lambda(\mathbf{x})\big)^{(1-y)}$$

$$\lambda(\mathbf{x}) = \text{sigmoid}(\mathbf{w}^{\top}\mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^{\top}\mathbf{x}}}$$

Logistic Regression

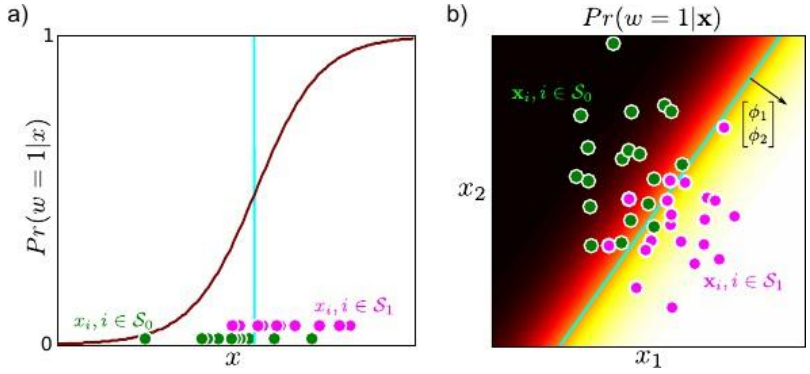
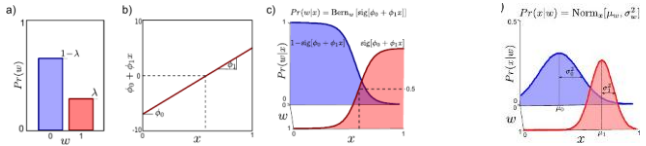


Figure from Prince (Ch. 9)

Logistic Regression vs Gaussian Classifier



Figures from Prince (Ch. 9)

Different learning:

- Gaussians: generative model, optimize $p(\boldsymbol{x}|y_0)$ and $p(\boldsymbol{x}|y_1)$
- Logistic Regression: discriminative model, optimize $p(y_1|\boldsymbol{x})$

Logistic Regression: MLE

Learning: maximize $\mathrm{p}(y|\boldsymbol{x})$ (discriminative method)

$$\mathrm{p}(\, y \mid \mathbf{x}, \mathbf{w}\,) = \prod_{i=1}^{N} \lambda(\mathbf{x_i})^{y_i} \big(1 - \lambda(\mathbf{x_i})\big)^{(1-y_i)}$$

$$\log \mathrm{p}(\, y \mid \mathbf{x}, \mathbf{w}\,) = \sum_{i=1}^{N} \big[y_i \log \lambda\,(\mathbf{x_i}) + (1 - y_i) \log\big(1 - \lambda(\mathbf{x_i})\big)\big]$$

$$= \sum_{i=1}^{N} \big[y_i \log \mathrm{sig}\,(\mathbf{w}^{\mathsf{T}}\mathbf{x_i}) + (1 - y_i) \log\big(1 - \mathrm{sig}(\mathbf{w}^{\mathsf{T}}\mathbf{x_i})\big)\big]$$

Logistic Regression: MLE

Learning: maximize $\mathrm{p}(y|\boldsymbol{x})$ (discriminative method)

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$$= \sum_{i=1}^{N} \big[y_i \log \mathrm{sig}\,(\mathbf{w}^{\top}\mathbf{x_i}) + (1 - y_i) \log\big(1 - \mathrm{sig}(\mathbf{w}^{\top}\mathbf{x_i})\big)\big]$$

Optimize by setting: no close form solution! Use gradient descent

$$\frac{d}{d\mathbf{w}} \log \mathrm{p}(\, y \mid \mathbf{x}, \mathbf{w}\,) = \sum_{i=1}^{N} \big(y_i - \mathrm{sig}(\mathbf{w}^{\top}\mathbf{x_i})\big)\mathbf{x_i} = 0$$

Hints: derivatives of sigmoid

$$\text{sig}(\mathbf{w}^\top \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^\top \mathbf{x}}}$$

$$\frac{d}{d\mathbf{w}} \text{sig}(\mathbf{w}^\top \mathbf{x}) = \text{sig}(\mathbf{w}^\top \mathbf{x})\big(1 - \text{sig}(\mathbf{w}^\top \mathbf{x})\big)\mathbf{x}$$

$$\frac{d}{d\mathbf{w}} \log\big(\text{sig}(\mathbf{w}^\top \mathbf{x})\big) = \frac{\text{sig}(\mathbf{w}^\top \mathbf{x})\big(1 - \text{sig}(\mathbf{w}^\top \mathbf{x})\big)}{\text{sig}(\mathbf{w}^\top \mathbf{x})}\mathbf{x} = \big(1 - \text{sig}(\mathbf{w}^\top \mathbf{x})\big)\mathbf{x}$$

$$\frac{d}{d\mathbf{w}} \log\big(1 - \text{sig}(\mathbf{w}^\top \mathbf{x})\big) = \frac{-\text{sig}(\mathbf{w}^{\top \mathbf{x}})\big(1 - \text{sig}(\mathbf{w}^\top \mathbf{x})\big)}{1 - \text{sig}(\mathbf{w}^\top \mathbf{x})}\mathbf{x} = -\text{sig}(\mathbf{w}^\top \mathbf{x})\mathbf{x}$$

Logistic Regression vs Conditional Gaussian

Number of parameters (D dimensions, 2 classes):

Gaussian distributions (equal priors)

$2 \times D$	(mean vectors)
$D(D+1)/2$	(shared covariance)
$D(D+5)/2$	(total, quadratic in D)

Logistic Regression

D (weights)

Training:

Gaussian distributions

- closed form solution
- generative model

Logistic Regression

- gradient descent
- discriminative model

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